NATIONAL INSTITUTE OF TECHNOLOGY, PATNA
MID SEMESTER EXAMINATION, March 2025

Program: B. Tech. [CSE (UG and DD)] Semester: 2 Department: Physics
Course Code: PH24101 Course Name: Engineering Physics
Full Marks: 30 Duration of Examination: 2 hours

Answer all questions. Please assume missing data suitably, if any.

Q1. (a) Discuss the theory of forced oscillations. How does the sharpness of resonance depend on damping? On what factors does the damping depend? 7+3

(b) A 4 kg mass is hung on the end of a helical string and is pulled down and let go to vibrate vertically. The mass completes 100 vibrations in 55 seconds. Calculate the force constant to the spring.

Q2. (a) Find the constant a, b, c such that the vector field 5+5
$\mathbf{F} = (x+2y+az)\,\mathbf{i} + (bx-3y-2)\,\mathbf{j} + (4x+cy+2z)\,\mathbf{k}$ is irrotational. Describe electrostatic boundary conditions for electric field vectors across dielectric mediums.

(b) Derive an expression for the em wave propagation in the conducting medium. A point charge of 12C is located at the center of a cube of side 5m. Calculate the electric flux through each of the surfaces.

Q3. (a) What do you mean by (i) Population inversion and (ii) Active system? 5+5
Find the energy stored in a one-meter length of LASER beam operating at 1mW.

(b) Explain the concept of Maxwell's displacement current and show how it led to the modification of Ampere's law.